Ericsson PE (IND-PE-03) Routing Protocol Flap Resolution SOP

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_4198

Description

This Standard Operating Procedure (SOP) outlines the corrective actions required to mitigate the consequences of a Routing Protocol Flap (RPF) failure on Ericsson's PE (IND-PE-03) node at the Indiranagar Exchange site. The failure domain in question refers to a BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, triggered by a Critical alarm code (ALARM_4198). The underlying protocol mechanics driving this failure can be attributed to the complex interactions between the OSPF and ISIS routing protocols, which are responsible for establishing and maintaining routing adjacencies between nodes. On Ericsson's PE equipment, the OSPF protocol uses the OSPF database to store routing information, which is then used to establish adjacencies with neighboring nodes. ISIS, on the other hand, uses its Link-State Protocol (LSP) to flood routing information throughout the network. In the event of a RPF, the PE node's OSPF and ISIS protocols may become desynchronized, leading to a loss of routing adjacencies and subsequent flooding of LSPs. The root cause of the RPF can be attributed to a variety of factors, including misconfigured routing protocols, hardware failures, or software bugs.

Pre-requisite Checks

- Check: Verify that the Indiranagar Exchange site is accessible and that the dispatch team has been notified to perform any necessary remote hands.

- Check: Confirm the status of the redundant link at the Indiranagar Exchange site to ensure that it is functioning correctly.

Potential Root Causes

Cause	Probability
Misconfigured OSPF protocol parameters	High
Hardware failure in the PE node's routing engine	Medium
Software bug in the PE node's routing protocols	Low

Diagnostic Commands & Expected Outputs

```
Command: show ospf database, Expected Output: OSPF database output indicating adjacency loss and routing information inconsistencies, Description: This command is used to verify the OSPF database and assess the impact of the RPF on the PE node's routing information.

Command: show isis database, Expected Output: ISIS database output indicating LSP flooding and adjacency loss, Description: This command is used to verify the ISIS database and assess the impact of the RPF
```

```
on the PE node's LSP flooding and adjacency loss.
```

Resolution Steps

Step 1: Step: Disable OSPF protocol on the PE node, Command: configure terminal router ospf no router ospf, Description: Disabling the OSPF protocol on the PE node will prevent further OSPF adjacency loss and allow the PE node to recover from the RPF.

Step 2: Step: Verify OSPF adjacency status, Command: show ospf neighbor, Description: This command is used to verify the OSPF adjacency status on the PE node and ensure that it has recovered from the RPF.

Step 3: Step: Clear ISIS database, Command: clear isis database, Description: Clearing the ISIS database will prevent further LSP flooding and allow the PE node to recover from the RPF.

Post-Check Verification

```
Command: show ospf database, Expected Output: OSPF database output indicating recovered routing information and adjacencies, Description: This command is used to verify the OSPF database and confirm that the PE node has recovered from the RPF.

Command: show isis database, Expected Output: ISIS database output indicating recovered LSP flooding and adjacencies, Description: This command is used to verify the ISIS database and confirm that the PE node has recovered from the RPF.
```

Roll-back Procedures

- Step: Re-enable OSPF protocol on the PE node, Command: configure terminal router ospf router ospf, Description: Re-enabling the OSPF protocol on the PE node will restore the original OSPF configuration and allow the PE node to resume normal operations.

- Step: Re-clear ISIS database, Command: clear isis database, Description: Re-clearing the ISIS database will restore the original ISIS configuration and allow the PE node to resume normal operations.